Linear Algebra lesson 1: Notation, Proofs

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I thought for our first math correspondence, we can start with some foundations for mathematical proof-writing. This was also how my linear algebra course started, and I found it very useful for my mathematical thinking. Thus, these concepts are not linear algebra-specific, but can be used for math proofs in general. See section 5 for a reference for notation and abbreviations, and the end for some practice exercises.

1. **Definitions**
   Here are some definitions to start with.
   (a) $\mathbb{N} = 1, 2, 3, 4, ...$ = set of natural numbers (can be thought of as positive integers)
   (b) $\mathbb{Z} = ..., -2, -1, 0, 1, 2, 3, ...$ = set of integers
   (c) $\mathbb{Q} = \{\frac{a}{b}$ such that $a, b \in \mathbb{Z}$ and $b \neq 0\}$ = set of rational numbers, which can be thought of any number that can be written as a fraction
   (d) $\mathbb{R}$ = set of real numbers, or any number you can typically think of. (It includes decimals, irrational numbers such as pi, and so forth.)
   (e) $\mathbb{C} = \{a + bi$ where a and b are real numbers and $i = \sqrt{-1}\}$ = set of complex numbers, or imaginary numbers.

2. **Induction**
   Induction is one form of logical reasoning that can be used for math proofs, usually when you are trying to prove something is true for all natural numbers ($\mathbb{N}$) or nonnegative integers. There are three main components to a proof by induction (base case, inductive hypothesis, and the inductive step). Let me illustrate using an example. Consider the statement

   $$1 + 2 + 3 + ... + n = \frac{(n)(n+1)}{2} \text{ for all } (\forall)\ n \in \mathbb{N}.$$

   For example, take $n = 5$:

   $$1 + 2 + 3 + 4 + 5 = 15 = \frac{5 * 6}{2}. \text{ It works!}$$

However, we need to prove this is true for any natural number, not just 5. We do so using the 3 steps of induction: 1) verifying the base case, 2) assuming the inductive hypothesis, 3) using the hypothesis to prove the statement with induction.

(a) Base Case ($n = n_0$ where typically $n_0 = 0$ or $n_0 = 1$)
This is the smallest value of n in the statement you are trying to prove. In our example, it is $n = 1$. So, just verify that it is true.

$$\text{For } n = 1, 1 = \frac{1 * 2}{2} = 1 \text{ is true.}$$

(b) Inductive Hypothesis (n=k)
Next, assume the statement is true for $n = k$, where $n \geq n_0$. In our example, we would write

$$\text{Assume for } n = k \text{ where } k \geq 1,$$

$$1 + 2 + 3 + ... + k = \frac{(k)(k+1)}{2} \text{ is true.}$$

(c) Inductive step ($n = k + 1$)
Now, we use our assumed hypothesis to prove that the case $n = k+1$ is also true. If we can prove this, then we have proven the statement is true for all positive integers. This is because you can inductively set k from 1 to progressively larger numbers, as k is arbitrarily chosen. So, in our example, we would do the following:

$$\text{Consider } n = k + 1.$$

$$1 + 2 + 3 + ... + k + (k+1) = (1 + 2 + 3 + ... + k) + (k+1)$$

$$= \frac{(k)(k+1)}{2} + (k+1) \text{ by our inductive hypothesis}$$

$$= \frac{(k^2 + k)}{2} + \frac{2 * (k+1)}{2} = \frac{(k^2 + k)}{2} + \frac{2k+2}{2}$$

$$= \frac{(k^2 + k + 2k + 2)}{2} = \frac{(k^2 + 3k + 2)}{2}$$

$$= \frac{(k+1)(k+2)}{2} \text{ by factoring}$$

$$= \frac{(k+1)((k+1)+1}{2}$$

$$= \frac{(n)(n+1)}{2} \text{ for } n = k + 1$$

Thus, we have proven

$$1 + 2 + 3 + ... + n = \frac{(n)(n+1)}{2} \text{ for all } (\forall) n \in \mathbb{N}$$

by the principles of mathematical induction.

This is how proof by induction works!

3. **Proof by Contrapositive**
All math proofs can be boiled down to essentially proving one statement implies another statement. Conceptually, we are just proving that $P \Rightarrow Q$ (where we use P to represent statement 1, $\Rightarrow$ to mean "implies," and Q to represent statement 2). However, sometimes, you can prove $P \Rightarrow Q$ in a roundabout way. More specifically, if we show when statement Q is false that shows that statement P is false, we have actually proven P implies Q. Below, we use the symbol $\neg$ to mean "not," or "false." I thought this was really interesting when I first learned about it.

   To summarize, proof by contrapositive is the following:

   (a) We want to prove $P \Rightarrow Q$. ("P implies Q.")
   (b) To do so, we show $\neg Q \Rightarrow \neg P$. ("not Q implies not P.")
      i. Thus, we assume the premise of $\neg Q$, or that Q is false.
      ii. Then we aim to arrive at the conclusion of $\neg P$, or that P is false.
   (c) This proves $P \Rightarrow Q$. Cool, right?

4. **Proof by Contradiction**
This is somewhat similar to proof by contrapositive. In this case, we want to prove $P \Rightarrow Q$ by showing that P and $\neg Q$ cannot both be our starting premise, as they lead to some contradiction.

   Proof by contradiction works as follows.

   (a) We want to prove $P \Rightarrow Q$. ("P implies Q.")
   (b) To do so, we show $\neg Q \quad \& \quad P \Rightarrow$ Contradiction. ("not Q given P implies contradiction.")
      i. Thus, we assume the premise of $\neg Q$, or that Q is false. We also assume the premise of P (which we will aim to contradict).
      ii. Then we aim to arrive at the conclusion that given $\neg Q$, P cannot also be true, else there exists some logical contradiction.
   (c) This proves $P \Rightarrow Q$.

5. **Notation** In math, there is a lot of notation. Here is a list of some common notations. If you ever encounter notation you don't understand, always feel free to ask in a letter. However, don't feel like you need to memorize these. You can always just write out the words instead of using notation shorthands.

   (a) : "such that"

(b) $\Rightarrow$ "implies"

(c) $\Leftrightarrow$ "equivalent"

(d) $\neg$ "not"

(e) $\in$ "in"

(f) $\exists$ "there exists"

(g) $\forall$ "for all"

(h) $\bigwedge$ "and"

(i) $\vee$ "or"

(j) $\therefore$ "therefore"

(k) AFSOC "Assume for the sake of contradiction" (often used at the start of proofs by contradiction)

(l) QED "The proof is finished."

6. **Exercises**
Below are some exercises that you can try as practice. I will focus on induction for this letter, and we can review proof by contrapositive and proof by contradiction in future correspondence.

(a) Using induction, prove that for all ($\forall$) $n \in \mathbb{N}, 1^2 + 2^2 + 3^2 + ... + n^2 = \frac{(n)(n+1)(2n+1)}{6}$.

(b) Using induction, prove that $\forall\ n \in \mathbb{N},\ n < 2^n$.

(c) Using induction, prove that $\forall\ n \in \mathbb{N},\ 2^{2n} - 1$ is divisible by 3.